Efficient Attention Mechanisms for Low-Latency Transformer Architectures

Authors: Dr. Emily Chen, Dr. Marcus Vance, AI Research Institute

Abstract: Large language models require quadratic time complexity in standard self-attention.
In this paper, we propose Linear Sparse Attention (LSA), reducing computational overhead by 60%.
Methodology: We partition attention heads into local dense and global sparse components.
Experiments: Tested on Wikitext-103 and CommonCrawl benchmarks across 8 NVIDIA A100 GPUs.
Results: Achieved 1.4x faster inference with minimal perplexity degradation (< 0.2).
Conclusion: LSA enables deployment of 70B parameter models on constrained edge hardware.